Moaaz Emam Ahmed

+20 1156637443 | moaaz.emam06@eng-st.cu.edu.eg | www.linkedin.com/in/moaaz-emam-8206cai
https://github.com/MoaazEmam

EDUCATION

Cairo University || GPA: 3.58 2022-2027
Major: Communication and Computer Engineering

SKILLS

- **Programming Languages:** C++, C, C#, Python, JavaScript, SQL, ARM assembly, Verilog HDL, Java
- **Development Tools & Environments:** Git, GitHub, Linux, Visual studio, IntelliJ IDEA, SQL Server, MongoDB, NodeJS, Postman, .NET framework, ModelSim, Logisim, Apache Airflow, Docker, Scrapy, Playwright, n8n, Airtable
- **Soft Skills:** Problem solving, Team collaboration, Rapid adaptability and learning, Effective communication

EXPERIENCE AND PROJECTS

Data Engineering Intern: *Instacodigo (Remote)* Jul 2025 – Oct 2025

- Designed and implemented an automated data pipeline using Apache Airflow for large-scale web data collection, processing, validation, and integration with Odoo ERP.
- Built a modular scraping framework using Scrapy and Playwright to extract structured data from static and JavaScript-rendered websites at scale.
- Implemented data cleaning, deduplication, serialization, and schema enforcement using Pydantic, with branching DAG workflows routing products, jobs, and articles to appropriate Odoo models via OdooRPC.
- Developed an automated social media workflow using n8n and Airtable, supporting scheduled content verification and posting to 13+ platforms with OAuth2 authentication and token refreshing.

Hospital Information Website: *Backend development course* 2025

- Developed and implemented a scalable hospital management system using Node.js, Express and MongoDB Atlas, allowing ease of access for patients, doctors and admins.
- Developed a secure authentication system using JWT and role-based access control for different user types, ensure privacy and secure access to sensitive information.
- Conducted unit and integration testing using Jest and Supertest to validate API functionality and system performance.

Google Docs Clone: *Advanced programming techniques* 2025

- Developed a real-time collaborative text editor in Java using Spring Boot, WebSockets, and JavaFX, supporting multi-user editing with role-based access.
- Implemented concurrent edits using a custom CRDT algorithm, multithreading, and asynchronous updates.
- Integrated features like import/export files, user tracking with colored cursors using JavaFX, user comments, and graceful disconnection and reconnection.

Fashion House Management System: *Databases* 2024

- Designed and implemented a multi-user windows application using C# and .NET framework, implementing event-driven programming and interactive GUI.
- Engineered a secure SQL relational database in SQL Server Management studio, designing ER diagrams and optimizing table relationships for efficient data retrieval and improved application responsiveness.
- Developed secure authentication system with hashed passwords and encrypted payment details to ensure safe card storage and transaction processing.

Custom Programming Language Compiler: *Compilers* 2025

- Designed and implemented a full compiler front-end for a custom programming language using Flex and Bison in C.
- Built lexical analysis, parsing, AST construction, and semantic analysis supporting variables, functions, control flow, loops, scoping, and basic data types.

- Designed grammar rules, symbol table, and three-address code (TAC) intermediate representation.
- Implemented semantic checks including type mismatch, undeclared/redeclared identifiers, and function parameter mismatch, and developed a GUI-based code editor for compilation and visualization.

OS Scheduler and Memory Manager: *Operating systems* 2024

- Developed an OS-level scheduler in C on a Linux environment that selects among 4 implemented scheduling algorithms: SJF, Preemptive HPF, RR (with configurable quantum), Preemptive MLFQ.
- Engineered a process generator to simulate task arrivals and integrated intercross communication using message queues, signals, and shared memory, while employing forking to handle concurrent processes.
- Designed and implemented a Buddy Memory Allocation system using a tree structure with recursive splitting and merging for efficient dynamic memory management and minimized fragmentation.